

Independent Learning in Constrained Markov Potential Games

Philip Jordan

Anas Barakat

Niao He



Motivation

Multi-agent RL in Markov Potential Games with:

- **independent learning** for:
 - (a) scaling (breaking curse of multi-agents),
 - (b) privacy (no information sharing),
 - (c) avoid communication cost.
- common coupled **constraints**; e.g., collision avoidance in autonomous driving, or power constraints in signal transmission

Related Work

	centralized	independent
MPG	Nash-CA; [1]; $\mathcal{O}(\epsilon^{-3})$	Ind. PGA; [2]; $\mathcal{O}(\epsilon^{-5})$
CMPG	CA-CMPG; [3]; $\tilde{\mathcal{O}}(\epsilon^{-5})$	<i>Ours (iProxCMPG)</i> ; $\tilde{\mathcal{O}}(\epsilon^{-7})$

Problem Setting

Markov Game $\mathcal{G} = (\mathcal{S}, \mathcal{N}, \{\mathcal{A}_i, r_i\}_{i \in \mathcal{N}}, c, \alpha, \mu, P, \kappa)$

- shared state space $\mathcal{S}$
- players $\mathcal{N} = \{1, \dots, m\}$
- joint policy space

$$\Pi = \prod_{i \in \mathcal{N}} \Delta(\mathcal{A}_i)^{\mathcal{S}}$$

- indiv. value functions

$$V_{r_i}(\pi) := \mathbb{E}_{s \sim \mu} \left[\sum_{t=0}^T r_i(s_t, a_t) \mid s_0 = s \right]$$

- potential structure:

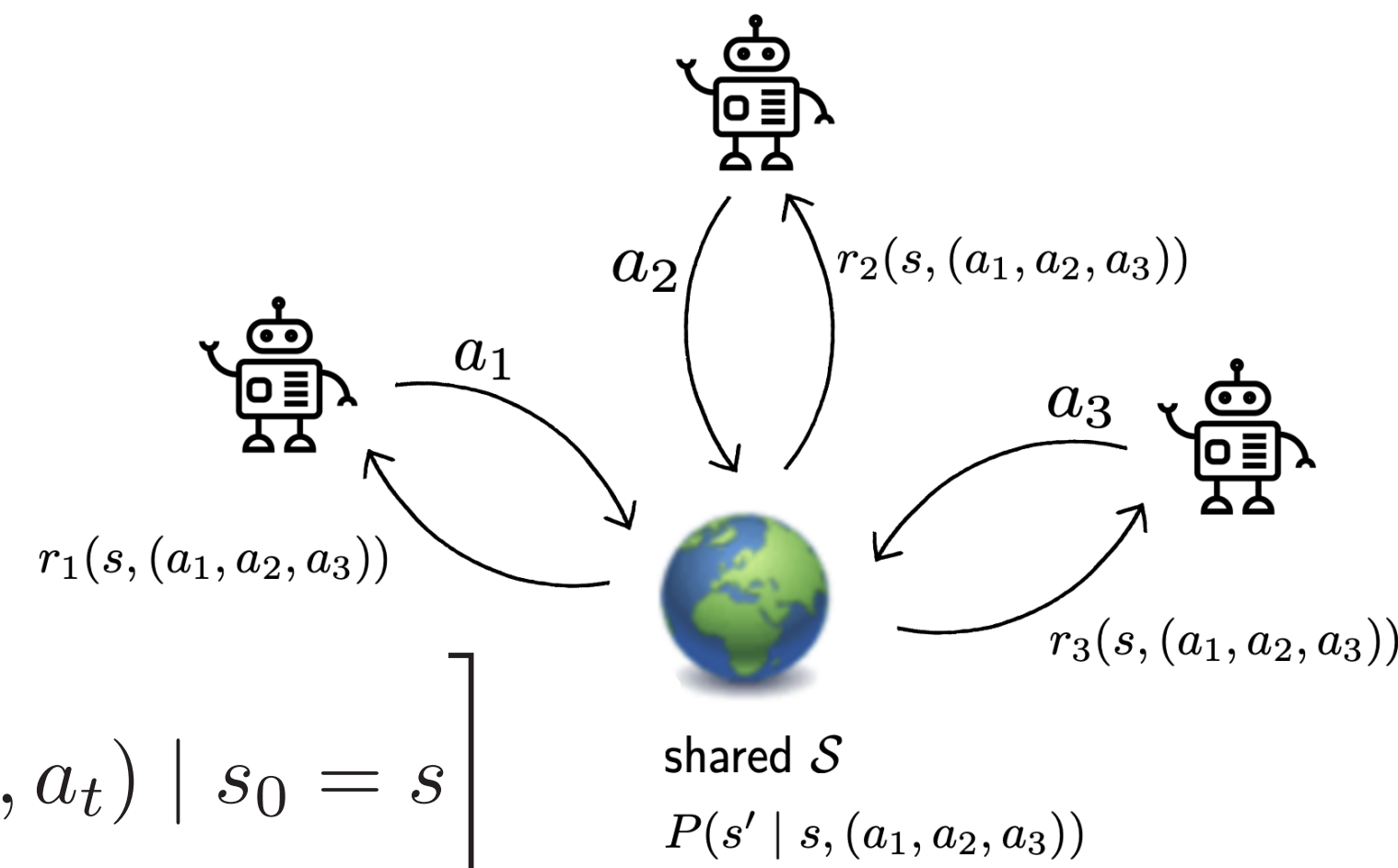
$$\exists \Phi : \Pi \rightarrow \mathbb{R} \text{ s.t. } \forall i \in \mathcal{N}, (\pi_i, \pi_{-i}) \in \Pi, \text{ and } \pi'_i \in \Pi'_i, \\ V_{r_i}(\pi_i, \pi_{-i}) - V_{r_i}(\pi'_i, \pi_{-i}) = \Phi(\pi_i, \pi_{-i}) - \Phi(\pi'_i, \pi_{-i})$$

- constr. threshold $\alpha \in \mathbb{R}$; feasible set $\Pi_c := \{\pi \in \Pi \mid V_c(\pi) \leq \alpha\}$,

$$V_c(\pi) := \mathbb{E}_{s \sim \mu} \left[\sum_{t=0}^T c(s_t, a_t) \mid s_0 = s \right]$$

- **solution concept**: ϵ -approx. NE: $\pi^* \in \Pi$ s.t. $\forall i \in \mathcal{N}, \pi'_i \in \Pi_c^i(\pi_{-i}^*),$

$$V_{r_i}(\pi^*) - V_{r_i}(\pi'_i, \pi_{-i}^*) \leq \epsilon.$$



Challenges

- nonconvex objective **and** constraint; constr. opt. challenge
- constraint **couples** π_i 's; how to learn independently?
- **no** strong duality [3]; prohibits CMDP primal-dual methods

Main Contributions

- design of an algorithm for **independent** learning of **constrained ϵ -approximate Nash equilibria** in CMPGs
- establish **sample complexity** with poly. dependency on ϵ and problem parameters
- two CMPG **applications**: pollution tax & energy marketplace

Method

- proximal-point-like update

$$\pi^{(t+1)} = \arg \min_{\pi \in \Pi} \left\{ \Phi(\pi) + \frac{1}{2\eta} \|\pi - \pi^{(t)}\|^2 \mid V_c(\pi) + \frac{1}{2\eta} \|\pi - \pi^{(t)}\|^2 \leq \alpha \right\}$$

converges to ϵ -KKT policy $\Rightarrow \epsilon$ -approx. constr. NE

- Φ and V_c weakly cvx $\Rightarrow$ subproblem obj. and constr. strongly cvx $\rightarrow$ solve via gradient switching

- **observation**:

$$\nabla_{\pi_i} \Phi_{\eta, \pi'}(\pi) = \nabla_{\pi_i} \Phi(\pi) + \frac{1}{\eta} (\pi_i - \pi'_i) = \nabla_{\pi_i} V_{r_i}(\pi) + \frac{1}{\eta} (\pi_i - \pi'_i) \\ \Rightarrow \text{implementable as independent PG steps}$$

Algorithm (iProxCMPG)

for $t = 0, \dots, T - 1$ **do**

$\pi_i^{(t,0)} = \pi_i^{(t)}$ **for** $i \in \mathcal{N}$

for $k = 0, \dots, K - 1$ **and** $i \in \mathcal{N}$ **simultaneously do**

$$\pi_i^{(t,k+1)} = \begin{cases} \mathcal{P}_{\Pi^i, \epsilon} \left[\pi_i^{(t,k)} - \nu_k \hat{\nabla}_{\pi_i} V_{r_i}(\pi^{(t,k)}) - \frac{\nu_k}{\eta} (\pi_i^{(t,k)} - \pi_i^{(t)}) \right] & \text{if } \hat{V}_c(\pi^{(t,k)}) + \beta - \alpha \leq \delta_k \\ \mathcal{P}_{\Pi^i, \epsilon} \left[\pi_i^{(t,k)} - \nu_k \hat{\nabla}_{\pi_i} V_c(\pi^{(t,k)}) - \frac{\nu_k}{\eta} (\pi_i^{(t,k)} - \pi_i^{(t)}) \right] & \text{otherwise} \end{cases}$$

$$\pi_i^{(t+1)} = \pi_i^{(t, \hat{k})} \text{ s.t. } \mathbb{P}(\hat{k} = k) = \left(\sum_{k \in \mathcal{B}^{(t)}} \rho_k \right)^{-1} \rho_k$$

output: $\pi_i^{(T)}$ **for** $i \in \mathcal{N}$

Convergence Results

Assumptions.

1. **initial feasibility**: $\pi^{(0)}$ satisfies $V_c(\pi^{(0)}) < \alpha$

2. **uniform Slater's condition**:

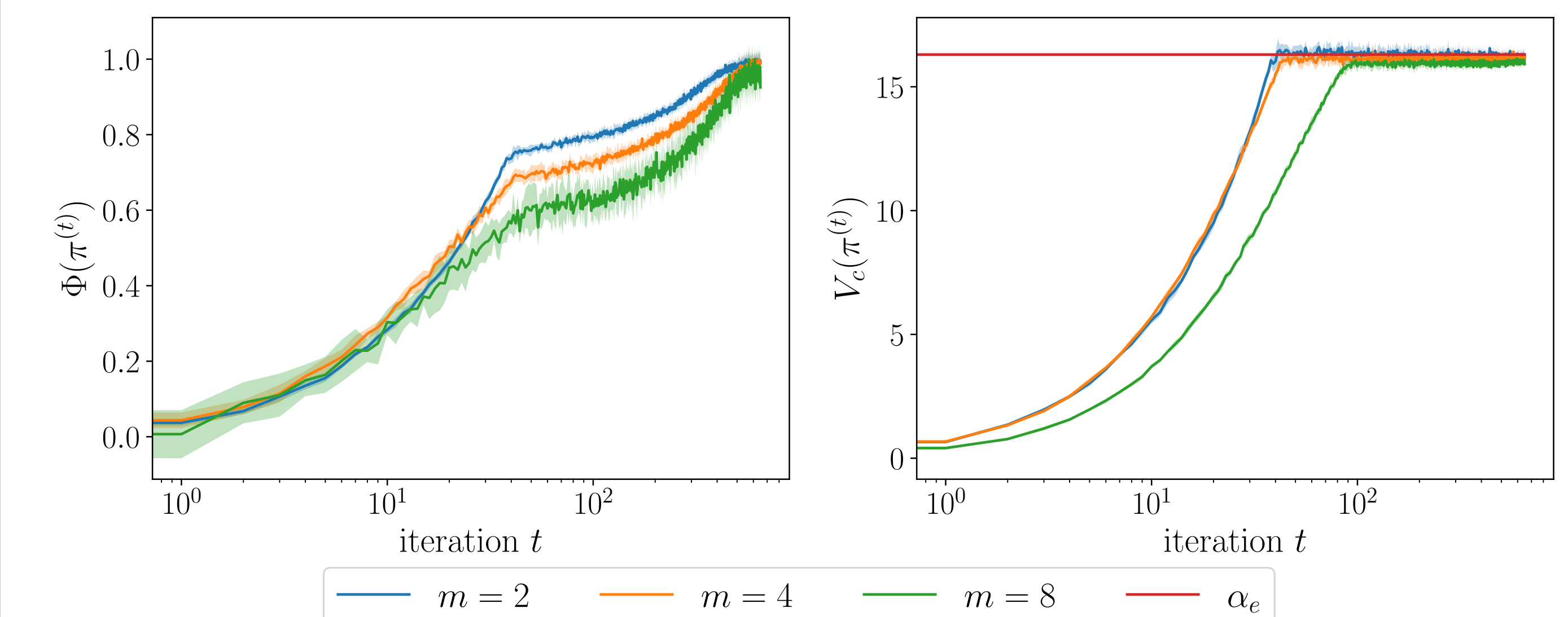
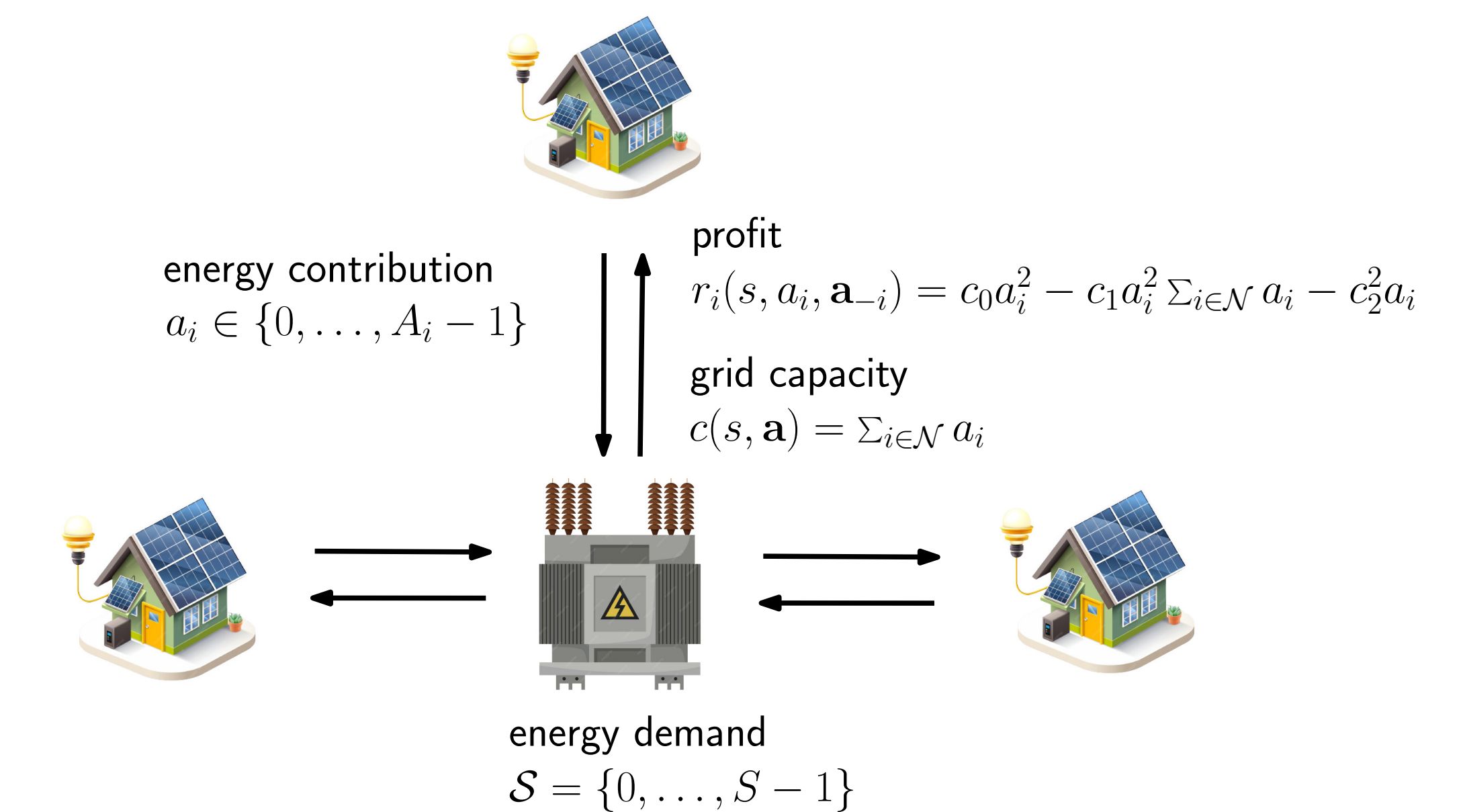
$$\exists \zeta > 0 \text{ s.t. } \forall \pi' \in \Pi \text{ with } V_c(\pi') < \alpha, \exists \pi \in \Pi \text{ s.t. } V_{\eta, \pi'}^c(\pi) \leq \alpha - \zeta$$

Theorem. For $\epsilon > 0$, using ϵ -greedy exploration, after running iProxCMPG for suitably chosen η, T, K , and $\{(\nu_k, \delta_k)\}_{0 \leq k \leq K}$, there exists $t \in [T]$ s.t. in expectation $\pi^{(t)}$ is a constrained ϵ -NE.

- **exact gradients**: total iteration complexity^a $\tilde{\mathcal{O}}(\epsilon^{-4})$
- **finite sample**: total sample complexity¹ $\tilde{\mathcal{O}}(\epsilon^{-7})$

^a $\tilde{\mathcal{O}}(\cdot)$ hides logarithmic dependencies in $1/\epsilon$, and polynomial dependencies in $m, S, A_{\max}, 1 - \gamma, \zeta$, and D .

Simulations: Energy Marketplace



Future Work

- learning constrained NEs beyond CMPGs
- “fully” independent learning (different stepsizes/algorithms)
- coupled playerwise (instead of common) constraints

References

- [1] Ziang Song, Song Mei, and Yu Bai. When can we learn general-sum markov games with a large number of players sample-efficiently? In *ICLR*, 2022.
- [2] Stefanos Leonardos, Will Overman, Ioannis Panageas, and Georgios Piliouras. Global convergence of multi-agent policy gradient in markov potential games. In *ICLR*, 2022.
- [3] Pragnya Alatur, Giorgia Ramponi, Niao He, and Andreas Krause. Provably learning nash policies in constrained markov potential games. In *AAMAS*, 2024.